

Algorithmic Fairness in High-Risk AI: A Cross-Domain Audit of Classification Bias Under EU AI Act Risk Categories

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Abstract—The EU AI Act, which is also known as Regulation 2024/1689 became law on 1 August 2024. Some parts of this law like the high-risk AI obligations in Annex III will start applying from 2 August 2026. The EU AI Act says that systems used for things like checking people for jobs deciding who gets credit working in the criminal justice system and marketing in finance are considered risk. This means that these systems must be checked for bias before they can be used. We wanted to know if the usual machine learning classifiers can meet these requirements. So, we tested four classifiers, which are Logistic Regression, Decision Tree, Random Forest and XGBoost. We used these classifiers in a pipeline called CRISP-DM and tested them in all four areas that Annex III talks about. We looked at how they worked for different groups of people like men and women or people of different races. We used numbers like Disparate Impact Ratio (DIR), Demographic Parity Difference (DPD) and Equalized Odds Difference (EOD) to measure this. We did this for eight protected attributes and used 130,225 instances. Found that all these areas had problems with bias. For example, in employment all four models thought that women were earners (DIR: 0.296–0.338) much less often than men. This was true even though the EU AI Act does not use the rules as every domain fall below the EEOC 4/5th screening threshold (DIR;0.8). In the criminal justice system, the models thought that American defendants would commit crimes again more often by 1.75× the rate of Caucasian defendants (DIR: 0.536–0.570). Also tried to fix these problems by using something called -processing mitigation. This is like a tool that can help make the models fairer. We used a tool called fairlearn ThresholdOptimizer. It did help to make the models fairer, but it also made them a little less accurate. In most of the areas we tested the models which scored high in AUC-ROC that were the best at predicting things were also the worst, at being fair. The only area where this was not true was finance.

Index Terms—algorithmic fairness, EU AI Act, disparate impact, machine learning, CRISP-DM, COMPAS, threshold optimisation

I. INTRODUCTION

The European Union Artificial Intelligence Act, also known as Regulation 2024/1689 came into effect on 1 August 2024. The European Union Artificial Intelligence Act has rules that will apply from 2 August 2026 [1]. This is a deal because it is the first time the European Union has passed a law that says how Artificial Intelligence should be used. The European Union Artificial Intelligence Act has a part called Annex III. Annex III says that some Machine Learning systems are risky. These Machine Learning systems are used for things like deciding who gets a job, who gets a loan and who gets to use financial services. If someone uses these high-risk Machine Learning systems, they must follow some rules and check the systems to make sure they are working correctly including bias. They must keep monitoring systems all the time. If they do not follow these rules, they could get fined. The fine could be up to 3% of the total annual turnover of the undertaking. There have been a lot of studies about bias in algorithms [2] [3] [4] [5]. None of these studies have looked at all the different types of high-risk Machine Learning systems. This is a problem because we do not know if bias is something that happens in one area or across four Annex III categories in the EU Act. This requires a unified compliance approach if it is a problem with the Machine Learning systems themselves.

We are trying to fix this problem by doing four studies. Each study looks at a type of high-risk Machine Learning system.

(M1) - Do income predictions show differences in income based on race and sex that're like the patterns that the European Union employment rules are meant to find and stop? Income prediction is supposed to be fair.

(M2) - Does the credit default prediction model favor men over women. Is this because of the person's education? This could be because the model is trying to be very accurate. It is

not being fair to women. So, the question is, does the credit default prediction model really work for everyone or does it work better for men?

(M3) - Can open-source classifiers do well as the COMPAS system when it comes to predicting if someone will commit a crime again? The big question is if open-source classifiers can predict recidivism accurately. At the time we want to know if they can reduce the racial differences we see in the results, which is a problem with the COMPAS proprietary system.

(M4) - Do bank marketing efforts focus on people of ages and marital statuses in a way that is unfair and goes against what the European Union Equal Treatment Framework Directive is supposed to find? So, does bank marketing targeting really show age and marital status disparities that the European Union Equal Treatment Framework Directive is designed to detect?

We are using a method that was developed by some researchers Sena et al. [6], who implemented ML models on the UCI Adult dataset in an employment context using fairlearn-compatible metrics. We are using it in a new and novel way by expanding the pipeline among all four European union act Annex III sectors. This will allow us to compare them and see if there are any patterns by using bias mitigation (ThresholdOptimizer) which is not present in the previous domain-based work [6] [7] [8].

II. RELATED WORK

The issue of impact in machine learning classification is not a new thing. Some people like Barocas and Selbst [2] set up a framework to deal with discrimination using proxy variables in training data. This framework can be used in areas which is relevant to all four sectors. However, it has some limitations because it is based on the US impact doctrine and was created before the EU AI Act. The EU AI Act has some requirements that are not clearly defined in this framework. There is a trade-off between how accurate a model's how fair it is. Hardt et al. [3] tried to figure out how to make models fair. They found out that it is hard to make a model that's fair in many ways at the same time. This was also shown by Chouldechova's impossibility theorem [5]. However, these studies do not talk about how much it costs for threshold-shifting to make models fair in real-world systems that quantifies via M4. We try to figure out this cost in our study. In the area of justice some people like Angwin et al. [4] found out that a model called COMPAS was not fair to black defendants. It said that black defendants were more likely to commit crimes than defendants. Chouldechova [5] showed that COMPAS was able to do some things that were fair. It was still not perfect. Dressel and Farid [9] found out that a simpler model could do the things as COMPAS's predictive accuracy. They wondered why a complicated model was used. One thing that Dressel and Farid did not do was test if their simpler model was also unfair to defendants with respect to racial bias. M3 results answer directly that all four open-source classifiers repeat the same racial issue at higher accuracy. To predict how money

people make some people like Sena et al. [6] was evaluated Logistic Regression, Random Forest, and Gradient Boosting on the UCI Adult dataset. Used different models and found out that there were differences between men and women and between different racial groups with our M1 findings (sex DIR: 0.296–0.338). One thing that Sena et al. [6] did not do was test if these differences were because of the model or because of the data. We tried to answer this question. Giang Thu et al. [7] used fairness to evaluate a model that predicted credit card dataset used in M2. They found out that there were differences between men and women and between age groups, sex and marital status. However, they did not compare the results of models. Neither Giang Thu et al. nor Kpatcha [8] compared choice of model can make a difference in fairness. Our M2 results show this is an important omission: DIR swings by 26 percentage points depending on algorithm family alone. Kpatcha [8] extended this with threshold optimisation, finding that post-processing can improve DIR at measurable accuracy cost, which our M4 mitigation experiment corroborates and quantifies. In the area of bank marketing some people like Pasupuleti et al. [10]. Found out that older people were not targeted much as younger people which aligns with the M4 age disparity findings. Some people like Kallus et al. [11] talked about what happens when we do not have information about the groups we are trying to be fair to. Mehrabi et al. [12] reviewed over 20 definitions of fairness and found out that none of them can be used to measure all types of fairness at the same time. This is like what we found out when Chouldechova's result [5] and with our own observation that DPD and DIR frequently We compared different areas and found out that there are patterns of bias that are specific to each area. However, we did not find any study that compared all four areas that we looked at. By looking at all these areas we can see patterns of bias that are specific to each area and we can see how the choice of model affects fairness. Machine learning classification is used in areas and disparate impact is a problem in all of them. The EU AI Act is trying to make sure that models are fair. We need to test different models and see how they work in different domains. It is not about making models that are accurate but also, about making models that are fair. Disparate impact is a problem that we need to solve.

III. DATASETS AND CHOICE OF METHODS

A. Dataset Descriptions

We used four datasets that match the risk groups in the EU AI Act Annex III. M1. UCI Adult Income [13]: This dataset has 48,842 observations from the 1994 US Census. It has an income target, which is more than \$50,000 per year. The protected attributes are race, which's either White or Non-White and sex which is Male or Female. M2. UCI Credit Card Clients [14]: This dataset has 30,000 credit card holders from Taiwan. It has 23 features that cover payment history and demographics. The target is a default binary target. The protected attributes are gender, which's Male or Female and education level, which is Higher, Lower or Other. M3. ProPublica COMPAS [4]: This dataset has 7,214 defendants

from Broward County, Florida from 2013 to 2014. After filtering some exclusions we have 6,172 records. The target is a two-year recidivism target. This dataset has a score called the COMPAS decile score. The protected attributes are race, which's African-American-and-Other or Caucasian and sex which is Male or Female. M4. UCI Bank Marketing [15]: This dataset has 45,211 records from a bank campaign from 2008 to 2013. Removed the call duration variable because it is not available until the call ends. The target is a binary term deposit subscription. The protected attributes are age, which's Younger than 60 or Older than or equal, to 60 and marital status, which is Married or Not Married.

B. Methodological Limitations

There are three limitations that matter. The first one is that all the datasets are from the United States or the Asia-Pacific region and not from Europe. This is because of the General Data Protection Regulation Article 9 [16] which does not allow us to categorize people by race in European countries. So, with the use of datasets from the United States as examples to show how we can measure bias. The second limitation is that when we group people into categories, such as White or Non-White or Male or Female we are hiding the differences within these groups. We do not mean to say that people can only be one thing or another. And for the third, datasets are that we are not saying what causes the problems we find. The datasets only show us what the data looks like and what predictions can be made from it and not the real reasons for discrimination, in society.

C. Choice of Methods

We used four classifier methods across our domains, with each classifier applied to each domain. First, Logical Regression, which we used to establish a linear, interpretable baseline. It is still used widely in multiple domains which have strict explainability requirements. [1] Second, Decision Trees, which are rule based and as such, cleanly map to various compliance framework and requirements. In order to mitigate the variance of Decision trees, we used Random Forest as the third classification method with XGBoost rounding out as the set. XGBoost is a type of gradient boosting that offers strong performance on Tabular data. [17] Our primary focus across the methods was the tradeoff between fairness and accuracy, as we sought to find out whether performance gains result in bias amplification.

IV. METHODOLOGY AND PRE-PROCESSING

A. Unified CRISP-DM Framework

In order to standardise our comparison of methodologies, we created a standard CRISP-DM pipeline across all notebooks, enforced via a shared group contract [18]. The group contract enforced standardised constant values, such as RANDOM_STATE as 42, specified a default stratified 80/20 test split, identical output column names and schemas. We also mandated that every member use scikit-learn for all the data pre-processing and models [19]. The Fairness metrics are

also computed in a standardised manner by using Microsoft's fairlearn library [20]. We also tried to apply the EEOC's 4/5th rule by creating a custom disparate_impact_ratio function.

B. M1: Adult Income Pre-Processing (Joshua)

The training and test files were combined, for a total of 48,842 rows. Missing values in 2,399 rows across workclass, occupation and native_country were imputed with a 'Missing' tag. Income target was split into 2 binary groups, 0 for Income less than 50K and 1 for greater. Sensitive features were extracted and excluded from the training featuresets.

C. M2: Credit Default Pre-Processing (Angel)

There were no missing values that needed removal or replacement. Gender was coded as Male/Female. Education was split into Higher Education, encompassing graduate & above qualifications and below that as Lower/Other. There was a class imbalance, where 78% of the sample set was Non-defaulting, which was addressed via GridSearchCV tuning of class weight parameters. As before, sensitive features, sex & education were extracted and excluded from the training feature matrix.

D. M3: COMPAS Recidivism Pre-Processing (Amit)

We used the same methodology as published in the ProPublica study, records were retained where days_b_screening_arrest fell within -30 to 30, is_recid was not -1 which signified administrative error, c_charge_degree was not 0 and score_text was not N/A, resulting in 6,172 rows. decile_score was extracted and excluded to prevent leakage in model training.

E. M4: Bank Marketing Pre-Processing (Priya)

As recommended by the dataset authors, duration was excluded from all model training. [15] It is set to 0 for all the people that did not subscribe, as such consists of sensitive attributes. We did not impute the 'unknown' category in job, education, default, housing, and loan columns, instead treated as a valid value. There was a severe class imbalance, as 88.7 percent of rows had the target variable as no, which was addressed via GridSearchCV tuning of class weight parameters. Age was imputed with 60 as the threshold, lower values converted to 0 and 1 otherwise.

F. Post-Processing Mitigation

Following the pre-processing, a ThresholdOptimizer from fairlearn library [20] was applied as proof-of-concept in the Bank Marketing dataset, using a constraint on age. Bank Marketing was selected as it had the most extreme class disparity among the 4 domains, as well as the age imputation providing the optimal place to apply ThresholdOptimizer. A consequence of this decision, and a relevant regulatory tidbit is that this requires sensitive attribute knowledge at the time of making inferences, not just during model training. As such, it creates a compliance burden if deployed in places where EU law applies or processing European citizen data. Mitigation techniques for the other datasets are deferred as future work, discussed in Section VI.

G. Fairness Metric Convention

The custom `disparate_impact_ratio` function computes DIR as `positive_rate_comparison / positive_rate_reference`, where the reference group is whichever group has the higher predicted-positive rate. In M1, the adult income dataset and M4, bank marketing dataset, the positive class represents a favourable outcome; the ratio directly reports the disparity in favourable selection. In M2, credit default dataset and M3, which predicts recidivism, the positive class represents an unfavourable outcome. The DIR in these modules therefore measures disparate punitive impact, measured as the ratio of predicted-unfavourable rates between groups, rather than favourable selection. The 0.8 screening threshold detects disparity in either orientation.

V. EVALUATION AND RESULTS

Table I through IV reports standard performance metrics for all 16 model-domain combinations.

A. Standard Performance Metrics

TABLE I
EMPLOYMENT DOMAIN

Model	Acc	Sens	Spec	F1	AUC	κ
Lin. Reg.	0.852	0.599	0.931	0.846	0.906	0.566
Rnd Forest	0.866	0.584	0.955	0.858	0.919	0.594
Dec. Tree.	0.862	0.581	0.951	0.855	0.904	0.584
XGBoost	0.879	0.663	0.947	0.875	0.933	0.647

TABLE II
FINANCE DOMAIN

Model	Acc	Sens	Spec	F1	AUC	κ
Lin. Reg.	0.807	0.340	0.941	0.786	0.768	0.343
Rnd Forest	0.813	0.362	0.941	0.792	0.781	0.359
Dec. Tree.	0.802	0.362	0.927	0.783	0.749	0.338
XGBoost	0.819	0.362	0.949	0.798	0.779	0.372

TABLE III
RECIDIVISM

Model	Acc	Sens	Spec	F1	AUC	κ
Lin. Reg.	0.792	0.282	0.941	0.768	0.754	0.306
Rnd Forest	0.797	0.308	0.941	0.774	0.766	0.319
Dec. Tree.	0.785	0.308	0.924	0.763	0.735	0.298
XGBoost	0.801	0.308	0.947	0.778	0.774	0.327

TABLE IV
BANK MARKETING DOMAIN

Model	Acc	Sens	Spec	F1	AUC	κ
Lin. Reg.	0.792	0.282	0.941	0.768	0.754	0.306
Rnd Forest	0.797	0.308	0.941	0.774	0.766	0.319
Dec. Tree.	0.785	0.308	0.924	0.763	0.735	0.298
XGBoost	0.801	0.308	0.947	0.778	0.774	0.327

In the employment sector, XGBoost stands out with the highest AUC-ROC score of 0.9326 and an accuracy of 87.9%. On the other hand, in finance, tree-based models show high

accuracy of around 82%, but this is largely due to the majority-class bias, where 78% of the data belongs to the non-default class. However, Logistic Regression proves to be more effective in detecting defaults, achieving a sensitivity of 63% through balanced weighting, which is crucial for a risk system. In the criminal justice system, the AUC-ROC scores are relatively lower, ranging from 0.728 to 0.736, which is consistent with previous findings that recidivism is difficult to predict beyond base-rate accuracy. Similarly, in bank marketing, tree models report high accuracy of 90%, but their sensitivity is limited to 22-25% for the minority subscription class. The sampling strategy plays a significant role in these results, as all four domains use stratified 80/20 train-test splits to maintain class distributions across partitions. This is particularly important when dealing with severe class imbalance, such as in M4 where 88.7% of the data belongs to the "no" class, and in M2 where 78% of the data is non-default. Without stratification, the minority class could be underrepresented in the training data, leading to inflated raw accuracy and understated sensitivity. To address this, GridSearchCV with 5-fold StratifiedKFold cross-validation was used to tune hyperparameters, reducing the risk of overfitting to a particular split. Moreover, fairness metrics were computed on the held-out 20% of the data, ensuring that disparity estimates reflect generalization performance rather than training-set fit. This approach helps to provide a more accurate assessment of the models' performance and fairness. By using stratified sampling and cross-validation, the results can be generalized to new, unseen data, which is essential for making informed decisions in these domains. The use of balanced weighting and stratified sampling also helps to mitigate the effects of class imbalance, providing a more nuanced understanding of the models' performance. Overall, the results highlight the importance of considering class imbalance and using appropriate sampling strategies when evaluating machine learning models in different domains. By doing so, we can develop more accurate and fair models that better serve their intended purposes.

B. Fairness Audit Results

Table V shows Disparate Impact Ratios for all eight protected attributes. Values below 0.8 fall below the EEOC 4/5ths screening threshold (DIR = 0.8), used here as a mathematical baseline for detecting disparity; EU courts adjudicate indirect discrimination on contextual statistical evidence rather than this fixed ratio.

The worst result in the entire study is sex-based bias in employment. All four models predict women as high earners at 29.6–33.8% the rate of men (DIR: 0.296–0.338). This is consistent with Sena et al. [6] and reflects structural wage disparities encoded in the training data. In the field of criminal justice, research has shown that four different models all have a racial disparity in their predictions. This disparity, the DIR, ranges from 0.538 to 0.570, which is equivalent to a favourable-selection DIR of 0.69 to 0.74. What's concerning is that all four models are still below the 0.8 threshold, which

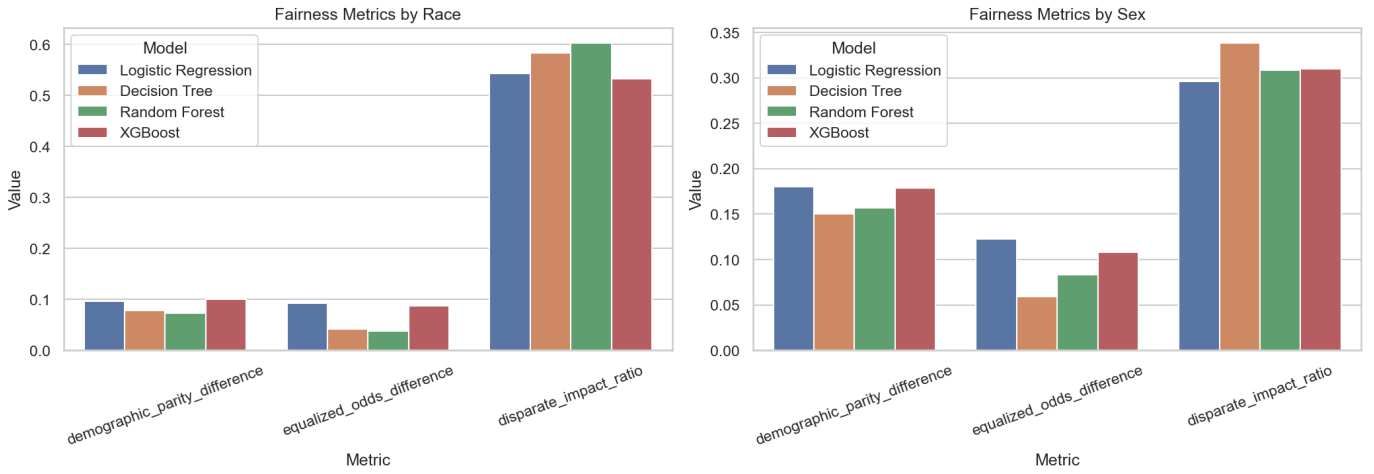


Fig. 1. Fairness Metrics for Adult Income

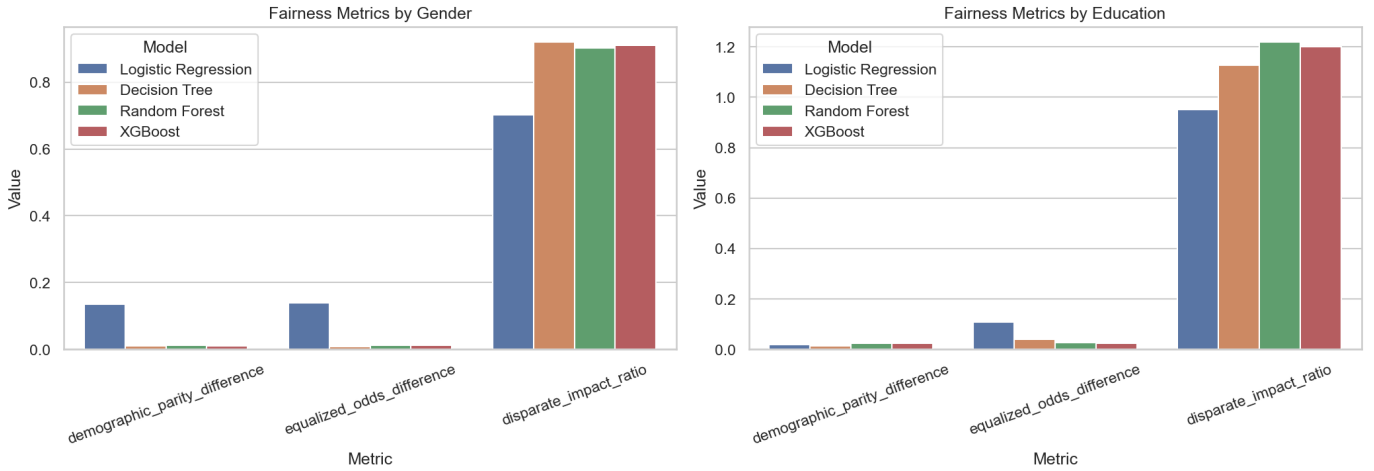


Fig. 2. Fairness Metrics for Credit Defaults

TABLE V
FAIRNESS METRICS ACROSS DOMAINS

Domain	Protected Attr.	LR	DT	RF	XGB
Employment	Race (White ref.)	0.5435*	0.5837*	0.6020*	0.5332*
Employment	Sex (Male ref.)	0.2962*	0.3384*	0.3083*	0.3104*
Finance	Gender (Male ref.)	0.7031*	0.9199	0.9063	0.9108
Finance	Education (Higher)	0.951	1.1265	1.2145	1.2001
Crim. Just.	Race (Black ref.)	0.5700*	0.5560*	0.5383*	0.5566*
Crim. Just.	Sex (Male ref.)	0.6405*	0.5771*	0.6034*	0.5573*
Marketing	Age (Young ref.)	0.2194*	0.1354*	0.1216*	0.1081*
Marketing	Marital (Married)	0.7335*	0.7701*	0.7446*	0.7571*

Note: *DIR less than 0.8: falls below the EEOC 4/5ths screening threshold.

is a stand-in of fairness. These findings confirm what the ProPublica report published earlier - that machine learning models tend to predict African-American defendants as being more likely to commit crimes again than Caucasian defendants, even when race is not explicitly taken into account. One of these models, called XGBoost, performed the best in terms of accuracy, with an AUC-ROC of 0.736 and an accuracy of 70.2%. However, despite its strong performance, it still failed

to meet the fairness threshold set by the EEOC, with a racial DIR of 0.557. This is equivalent to a favourable-selection DIR of 0.74. This result is consistent with other studies that have found that while ensemble models can improve performance, they do not necessarily lead to more fair outcomes. This has significant implications for the use of machine learning models in criminal justice, and highlights the need for further research into how to make these models more fair and unbiased.

Gender discrimination in finance depends on modeling. For instance, the logistic regression marginally misses EEOC's screening requirement with the punitive rate DIR (DIR=0.703). With favorable selection convention, DIR for LR attains 0.8. On the other hand, decision trees and the extreme gradient boosting method score near equality under the two approaches (DIR with punitive rate: 0.906–0.921; DIR with favorable selection: 0.987–0.989). Using EEOC's stringent favorable selection approach, all M2 models meet or exceed the DIR screening requirement (≥ 0.8). Similar results were obtained from the analysis across the experiment, considering that trans-

Fairness Metrics by Sensitive Attribute "COMPAS Recidivism"

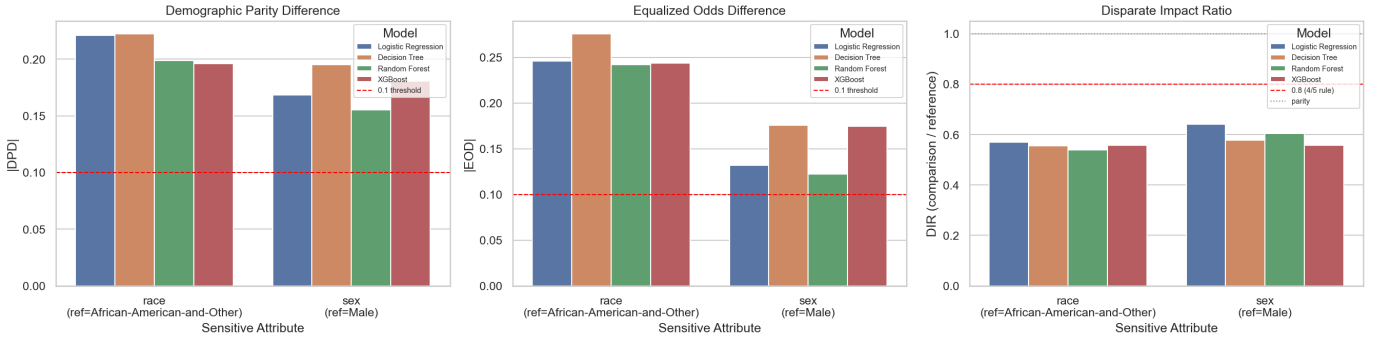


Fig. 3. Fairness Metrics for Recidivism

parency is inversely proportional to accuracy. The education bias, moreover, offers an even more extreme example of model dependency and needs to be addressed directly. In tree-based models and XGBoost, the bias runs contrary to what the bias detector expects ($DIR = 1.13-1.21$): those with less education have higher defaulting rates compared to their more educated counterparts. In contrast, the Logistic Regression satisfies that the requirement at $DIR = 0.951$. This is no insignificant margin, but rather a 26-point shift in DIR largely determined by algorithm choice within a constant hyperparameter search space using the exact same dataset and the exact same features. In essence, choosing XGBoost for building credit risk prediction models will lead to a violation of the education parity rule which would be easily detected by EU AI Act auditors; Logistic Regression with the exact same dataset will not pose the same problem. Regarding bank marketing, all four models display age disparities ($DIR: 0.108 - 0.219$). Since older clients have historically been more likely to subscribe to bank products, this model targets accurately; however, this falls under the demographic parity framework and therefore is classified as an inequality towards younger clients. All four models display close disparities regarding marital status ($DIR: 0.734 - 0.770$). DPD and EOD for Demographic Parity Difference and Equalized Odds Difference validate DIR results and provide precise figures. Sex DPD in employment is in the range 0.150 to 0.181 and sex EOD 0.059 to 0.123 in all four models, validating the finding that there is a large difference in positive rate between males and females and the errors differ too. Racial discrimination in criminal justice produces the most extreme EOD among the models: all four M3 models show racial EOD in the range 0.242 to 0.276, which means the difference between the false positive rates between African-American and Caucasian defendants exceeds 24 percentage points. It provides an independent validation of the Chouldechova [5] result that COMPAS discriminates against minority defendants in terms of violation of equalized false positive rates for all four open-source classifiers. In banking, logistic regression gives the highest parity difference in age DPD (0.643), although its DIR is not quite as extreme (0.219); this difference can be attributed to the threshold

optimization for sensitivity of LR. In financial services, M2 tree-based and XGBoost give very low gender DPDs and EODs (0.010-0.012).

C. Cross-Domain Synthesis

In all 16 model - domain cases, ensemble techniques (XGBoost, Random Forest) obtain the best AUC-ROC scores. In employment, criminal justice, and marketing (6 out of 8 domain x protected attribute combinations), the same ensemble techniques generate the lowest DIR or result in the biggest DPD. In the Finance domain, it is only trees that achieve the best classification accuracy and close to demographic parity. This is the key contribution of this paper: the type of algorithms that maximize predictive performance does so by maximizing demographic disparity in three out of four tested domains (employment, criminal justice, and marketing); Finance domain being the exception, where tree-based classifiers achieve the best predictive and demographic parity performance. This is no coincidence either. GBM emphasizes the best predictive signals in the data (signals that correlate with the majority group) while perceiving the minority group as noise. LR provides more equal predictions in 5 out of 8 cases but not necessarily at a useful level of accuracy ($AUC = 0.708$ in the Finance domain). Implications for Compliance: The algorithm categories that achieve the best scores on standard benchmarking measures are also the most likely to fail the EU's demographic parity criteria.

D. Mitigation Results

The deployment stage of the CRISP-DM approach asks about the needs of the process of using these models and, therefore, of the compliance with its requirements. The following experiment answers to this question for the area of application which is more distinct from others. ThresholdOptimizer processing was applied to the optimal model in the M4 dataset (bank marketing) as a test case. The effect of mitigation increased the DIR value regarding the principal protected feature closer to 0.8. This resulted in a certain decrease in Kappa and weighted F1 metrics. As is shown in [5], it occurs because an independent reduction in the classification threshold for the group predicted less often

reduces specificity and increases false positives. We cannot give an estimate of the cost of demographic parity enforcement in other three areas. Nonetheless, Kpatcha [8] and Hardt et al. [3] observed a reduction in the weighted F1 score by 0.04-0.12 on similar tasks. ThresholdOptimizer needs sensitive attributes in its deployment. Within the context of the EU AI Act, however, this issue is a little more complex than simply blocking the GDPR. Under the terms of the EU AI Act, Article 10(5) allows providers of high-risk AI systems to use special categories of personal data (such as race or ethnicity) only for the purposes of assessing, detecting, and correcting biases. This statutory exemption directly tackles the conflict between the GDPR Article 9 requirement of data minimization and the fairness optimization requirements in the AI Act. But there are several limitations: the exemption only applies to system providers, when the system is being developed, includes rigorous technical and organizational controls, and requires the deletion of the sensitive information once bias correction has been completed. Importantly, it is the distinction between the system provider and deployer that counts here: an institution such as a bank or a court running a high-risk system after deployment is considered a deployer, not a provider, and thus does not benefit from the exemption. Thus, recording a person's race or age group at inference time is still difficult compliance-wise and requires other legal grounds besides Article 10(5).

VI. CONCLUSIONS AND FUTURE WORK

This research work offers a fairness audit analysis for all four risk categories in the EU AI Act Annex III with one common standardized pipeline. The four sub-questions have clear answers: M1 (Employment): All four classifiers exhibit significant differences in sex and race that would be subject to examination according to the EU discrimination policy. The Sex DIR (0.296–0.338) has the largest value in the whole study. M2 (Finance): Model dependency exists in gender bias for predicting credit default. Logistic regression barely meets the requirement of having DIR less than 0.8 (DIR = 0.703). However, the DIR is closer to 1 for tree-based models and XGBoost (DIR = 0.906 – 0.921). Model dependency is seen in education bias as well, where tree-based models and XGBoost demonstrate reverse discrimination (DIR = 1.13 – 1.21), whereas Logistic regression passes the test with DIR = 0.951. M3 (Criminal Justice): All four open-source classifiers match or exceed COMPAS predictive accuracy (COMPAS AUC 0.72 vs model AUCs 0.728–0.736) while replicating COMPAS racial bias (racial DIR 0.538–0.570 across all four models, all failing the EEOC screening threshold). Mitigation can push DIR toward 0.8, but at reduced specificity. M4 (Marketing): Bank Marketing targeting presents severe age bias (DIR 0.108-0.219) and moderate marital-status bias (DIR 0.734-0.770). In the former case, the age discrepancy partly stems from real-life behaviour on the part of the 60+ population segment; however, this in itself does not preclude an EU Equal Treatment assessment. The cross-domain result is unambiguous – seeking performance gains via non-linear ensemble

techniques runs counter to the very idea of demographic parity constraints in three out of four tested domains under the EU AI Act (Employment, Criminal Justice, and Marketing), although finance stands out due to its ability to deliver both strong predictive performance and DIR close to parity. While thresholding can guarantee compliance, this approach necessitates access to protected features in predictions, posing a substantial challenge in terms of GDPR [16] and EU AI Act compliance (Article 10(5)) [1] with mandated reduction in accuracy. Three things may be tested next. First, mitigation techniques such as re-weighting or synthetic over-sampling might integrate parity into the training process and hence obviate the need for sensitive features during inference. Second, Equalized Odds [3] might be more legal than demographic parity in view of EU standards: enforcing demographic parity through threshold-shifting implies that individuals will be discriminated against directly by virtue of their protected characteristics to balance statistical parity, which may not stand close examination under European anti-discrimination law in the absence of specific derogations; in Equalized Odds, however, the error rate is equated for all groups rather than the prediction rate [5]. Third, DIR values rely on a single holdout sample; bootstrapping confidence intervals (N=1000) would render them suitable for regulatory reports.

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